

MAKING LIVES BETTER THROUGH EDUCATION

Scalability in Big Data Processing using Tableau

Presented by:

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Objectives:

At the end of the session, you will be able to:

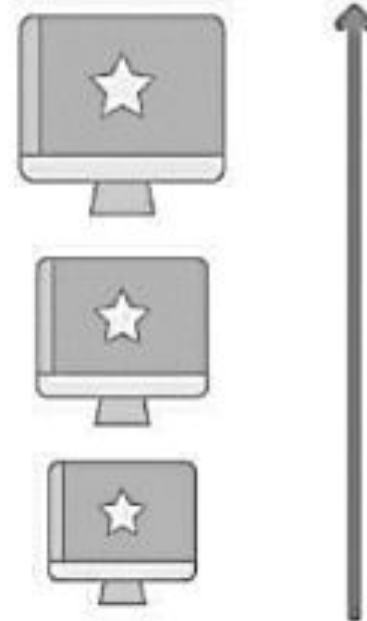
1. Differentiate Tableau Prep Builder and Tableau Desktop
2. Identify the features of the Tableau application
3. Perform Tableau Prep Builder installation



Scalability

- Scalability can be defined as the **capability of a system, network, or process to handle a growing amount of work or its potential to be enlarged to accommodate that growth.** Simply speaking, your application might be great now, but will it still be great when the whole organization wants to use it? Will it (and you) be able to cope?

VERTICAL SCALING
Increase size of instance
(RAM, CPU etc.)



HORIZONTAL SCALING
(Add more instances)



Big Data Scalability

- Big Data Scalability refers to the **ability of a system, framework, or technology to efficiently handle growing volumes of data, users, or processing demands without compromising performance.**

In simple terms, it means the system can “**scale up**” (add more powerful resources) or “**scale out**” (add more machines or nodes) as the amount of data and processing requirements increase.



Workload

- A system is considered scalable when it **doesn't need to be redesigned to maintain effective performance during or after a steep increase in workload.**
- refer to **simultaneous users, storage capacity, the maximum number of transactions handled**, or anything else that pushes the system past its original capacity.



Scalability in Big Data Processing using Tableau

- Scalability here means **Tableau's ability to efficiently analyze, visualize, and interact with very large datasets as the volume of data and number of users grow.**
- Instead of struggling with performance slowdowns when data scales to millions or billions of rows, Tableau provides mechanisms to adapt and optimize the processing of Big Data.



How Tableau Handles Big Data Scalability

Live Connections to Big Data Systems

- Tableau connects directly to databases like Hadoop, Spark, Google BigQuery, Amazon Redshift, and Snowflake.
- Heavy computations are pushed down to these systems, letting Tableau scale without importing all data locally.

Extracts (Tableau Data Extracts – TDE/Hyper)

- Tableau can create compressed extracts of massive datasets. These extracts are optimized for fast querying, aggregation, and visualization. Scales well for millions of rows without straining the source system.



How Tableau Handles Big Data Scalability

Parallel Query Processing

- Tableau can process queries in parallel, leveraging modern CPUs and distributed systems to maintain speed.

Incremental Refresh

- Instead of reloading the entire dataset, Tableau updates extracts incrementally, making it scalable for continuously growing Big Data sources.

Data Blending and Aggregation

- Tableau allows working with summarized/aggregated views (not the raw row-level detail) to improve performance while still gaining insights.



How Tableau Handles Big Data Scalability

Server & Cloud Scalability

- Tableau Server / Tableau Cloud scales horizontally by adding more nodes to handle growing users and workloads.

Example

- Imagine a telecom company with billions of call records:
- Tableau connects live to Hadoop/Spark for heavy computations (scalable live queries).
- It uses Hyper extracts for dashboards that need speed.
- Analysts can still filter, drill down, and visualize without waiting minutes for queries.
- If more teams join, Tableau Server can be scaled by adding more nodes to handle concurrent users.



Tableau Prep Builder

- Tableau Prep Builder is a **data preparation tool from Tableau that helps you clean, combine, and shape raw data before analyzing it in Tableau Desktop or Tableau Server.**
- Think of it as the step **before visualization** — it makes messy, scattered, or complex data ready for analysis.



Key Features of Tableau Prep Builder

- ✓ **Data Source Connection** – the application allows you to connect to multiple sources of data, such as but not limited to spreadsheets, databases, CSV files, and text files.
- ✓ **Data Cleaning and Transformation** - the application allows the cleaning of data by removing duplicated values, correcting misspelled words, or removing null values, uniformly formatting characters, and can do combining or splitting fields.
- ✓ **Drag and Drop** – you can drag and drop or double-click to bring your tables into the flow pane
- ✓ **Data Joining and Blending** - you can join multiple data sources together or blend data from different sources to create a single dataset.
- ✓ **Data Sampling** - you can take a sample of your data to preview the changes made during the data preparation process.
- ✓ **Data Output** - once you have prepared your data, you can output it in various formats such as Excel, CSV, and Tableau Data Extract (TDE) files.



Key Features of Tableau Prep Builder

Example Use Case

- Imagine you have:
 - Excel file of customer details
 - SQL database with transaction records
 - CSV with regional codes
- But the data has issues: duplicates, mismatched names, wrong formats.

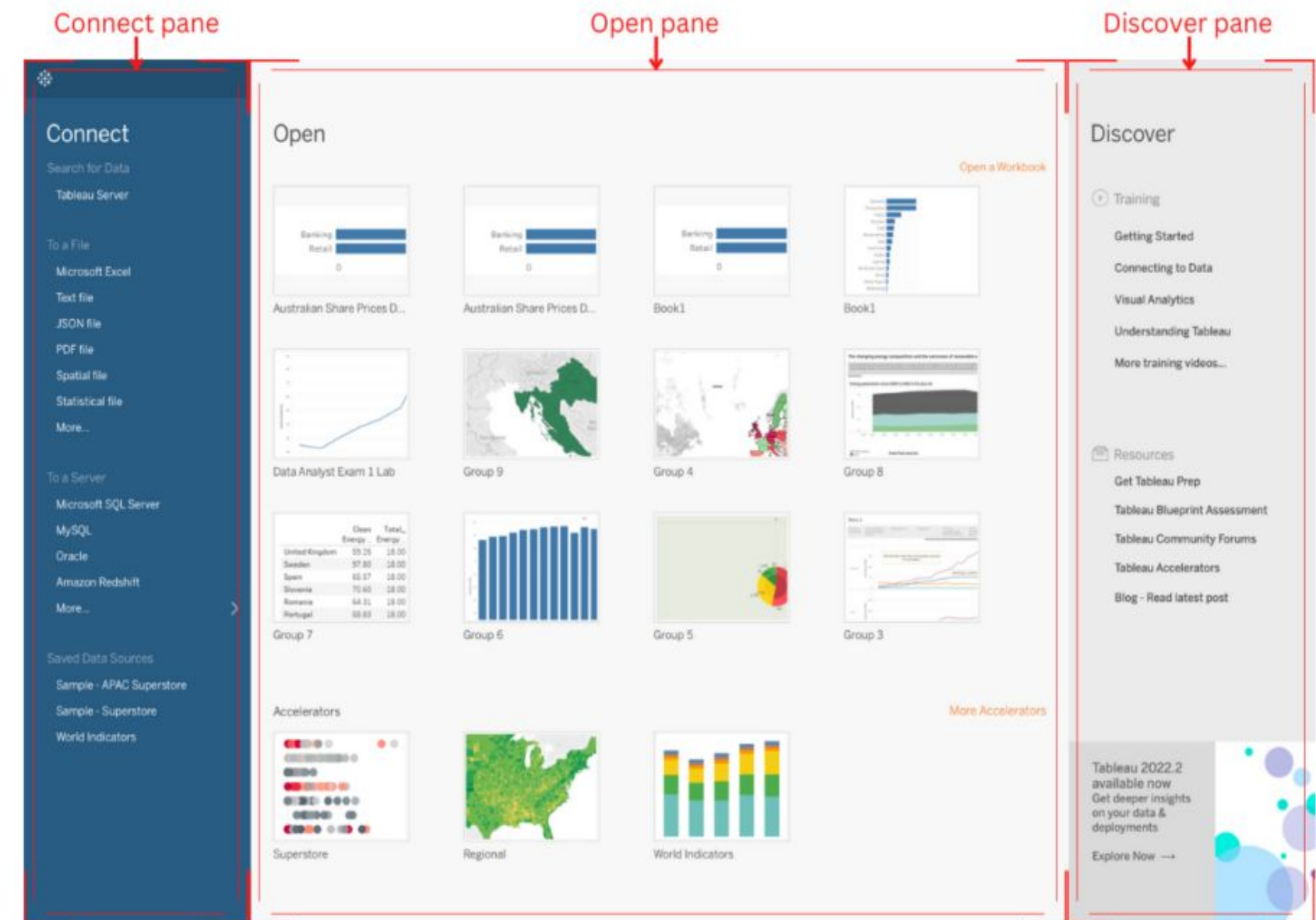
With Tableau Prep Builder you can:

- Join the customer + transaction data
- Clean spelling errors in customer names
- Standardize date formats
- Aggregate transactions by region
- Export the final, clean dataset as a Hyper Extract → ready to visualize in Tableau.



Tableau Desktop

- Tableau Desktop is the visualization tool for Tableau products. It is the primary authoring and visualization tool of Tableau. It allows analysts and users to **connect to data, create interactive dashboards, and perform advanced analytics through a drag-and-drop interface.**
- This is where you analyze data and build dashboards before sharing them through Tableau Server, Tableau Cloud, or Tableau Public.



Demonstration of Tableau



Skill-Building Activity (20PTS)

For this activity, you will download and install the Tableau Prep Builder installer into your laptop or computer unit.

Installing Tableau Prep Builder or Tableau online To use Tableau in preparing your datasets and setting connections, you may opt to download and install Tableau prep builder or use Tableau online.

Note:

- If your computer unit met the system requirements of Tableau Prep Builder, you may download the software from the link as indicated below. <https://www.tableau.com/products/prep/download>, Once you have downloaded the installer, double-click the executable file and just follow the simple on-screen prompt to finish the installation process.



Skill-Building Activity (20PTS)

- If your computer unit did not meet the requirements, use Tableau Online instead:

<https://www.tableau.com/products/online/confirmation>

1. Note that as a first-time user, you need to create and register an account.
2. After creating your account, you will receive an email to confirm your account
3. Click the link on your email to activate your account.
4. Create your Tableau Online site.



Check for Understanding (10PTS)

Based on the content notes, provide what is being asked in each number.

1. Based on the installation of Tableau Prep Builder, how many does will the trial version last?

2. TRUE/FALSE. Tableau Prep is mainly used for data visualization.

3. Give at least 2 system requirements to be able to use or run Tableau Prep Builder

4. TRUE/FALSE. Once installed, both Tableau Prep and Tableau Desktop can be used

offline. _____



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